12.10 Methods of Preparation of carboxylic acids

Some important methods for the preparation of carboxylic acids are as follows :

1. From Primary alcohols and aldehydes

Primary alcohols and aldehydes can easily be oxidised to the corresponding carboxylic acids with oxidising agents such as potassium permanganate (in acidic or alkaline medium), potassium dichromate (in acidic medium)

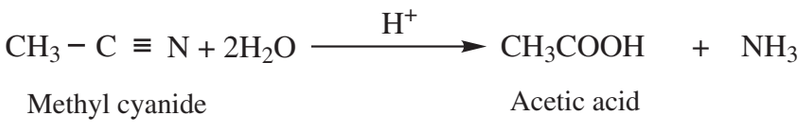
Example



2. Hydrolysis of Nitriles

Nitriles yield carboxylic acids when subjected to hydrolysis with an acid or alkali.

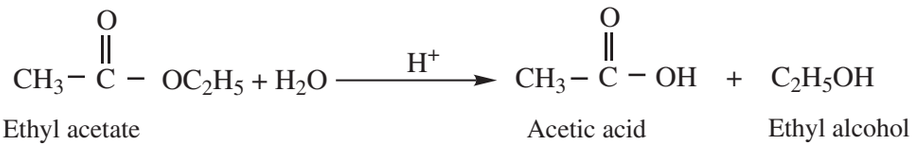
Example



3. Acidic hydrolysis of esters

Esters on hydrolysis with dilute mineral acids yield corresponding carboxylic acid

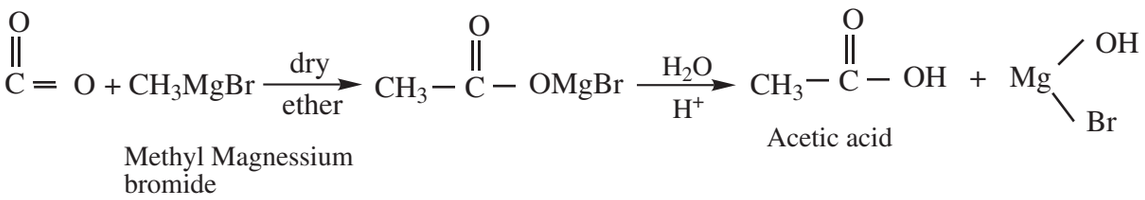
Example

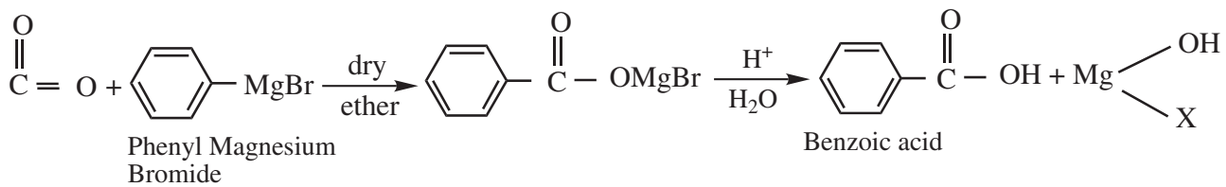


4. From Grignard reagent

Grignard reagent reacts with carbon di oxide (dry ice) to form salts of carboxylic acid which in turn give corresponding carboxylic acid after acidification with mineral acid.

Example





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